

AI-Powered Multimodal Medical Research Abstract Finder

Overview

Healthcare professionals and medical researchers often need to quickly identify relevant studies from a rapidly growing collection of medical publications. Traditional keyword-based search systems frequently fail to capture the true meaning of complex medical queries and may miss relevant research.

Researchers may search using natural language questions such as:

- *“Latest treatments for early-stage pancreatic cancer.”*
- *“Non-invasive therapy for knee arthritis.”*
- *“mRNA vaccine studies published after 2022.”*

In modern research workflows, information may also come from multiple formats such as clinical notes, screenshots of research figures, recorded discussions, or uploaded documents. Traditional search tools struggle to interpret these different types of inputs or connect them with relevant scientific literature.

Objective

The objective of this project is to develop an AI-powered multimodal medical research retrieval and analysis system that allows clinicians and researchers to explore medical publications using natural language queries and other forms of input such as documents, images, or voice queries.

The system should retrieve the most relevant medical research abstracts based on semantic similarity and provide structured insights, summarized findings, and supporting evidence to help users interpret research across multiple studies.

Key Capabilities

Multimodal Query Understanding

The system can interpret user queries provided through text, voice, or uploaded research documents.

Semantic Research Retrieval

Relevant medical studies are retrieved based on conceptual similarity rather than simple keyword matching.

Context-Aware Medical Interpretation

The system understands medical terminology, disease categories, and treatment types when analyzing queries.

Research Insight Summarization

Key findings across multiple research papers are summarized to help users quickly understand the overall evidence.

Research Filtering and Exploration

Users can filter results based on factors such as publication year, disease type, or study category.

Evidence-Based Results

Retrieved abstracts include supporting references to help researchers evaluate the reliability and relevance of the studies.

Requirement 1 (Basic)

- Basic RAG with medical abstract embeddings
- Semantic search across research papers
- Simple relevance ranking
- Medical terminology validation guardrails
- Citation extraction
- Metadata filtering (publication year, journal, study type)
- Query expansion using medical terminology (MeSH terms)

Expose the core functionality through API endpoint so that users or other applications can interact with the system.

Requirement 2 (Advanced)

- DeepEval with domain-specific medical metrics
- Custom evaluation: clinical relevance, evidence quality, study design assessment
- LLM-as-judge with medical knowledge validation
- Token usage tracking for literature review automation
- Performance testing: scalability across 36M+ abstracts
- Medical accuracy guardrails and fact-checking

Build a simple front-end interface to demonstrate how an end user can interact with the service.

Hybrid Research Retrieval

- Hybrid search combining vector similarity and keyword retrieval
- Reranking using cross-encoder models for improved research relevance
- Dynamic filtering by research attributes (disease area, publication year, clinical trial stage)

Medical Research Intelligence Layer

- Automated summarization of multiple research papers
- Identification of conflicting findings across studies
- Trend analysis for emerging treatments and therapies
- Knowledge graph generation connecting diseases, treatments, and outcomes

Multi-Agent Research Analysis System

Implement a specialized agent network for deeper research analysis:

- Research Retrieval Agent – retrieves relevant research abstracts
- Methodology Critic Agent – evaluates study design and methodology
- Statistical Reviewer Agent – analyzes statistical validity and significance
- Clinical Applicability Agent – assesses real-world medical relevance
- Research Summarization Agent – synthesizes insights across multiple studies

Additional Capabilities

- Handoff to full-text retrieval and meta-analysis agents
- Agent-to-Agent (A2A) communication for systematic review collaboration
- Automated literature review generation for complex medical queries

Deliverables

1. Architecture Diagram

Include a high-level view of your system in JPEG or PDF.

Show how data flows through:

- research data ingestion pipeline
- document chunking and embedding generation
- hybrid retrieval system
- multi-agent analysis layer
- final response generation

2. Design

Articulate system design decisions taken and the trade-offs made, such as:

- vector database selection
- chunking strategy for medical abstracts
- hybrid retrieval vs semantic-only retrieval
- agent orchestration design
- guardrail implementation for medical accuracy

3. Full Executable Code (Microservice)

Focus on clarity and modularity.

Provide a README explaining:

- Project setup (how to install or run)
- Data ingestion and indexing process
- Example research query and retrieved results
- Example of generated research insight or literature summary

4. Panel Presentation (10 minutes)

Demo of your working solution outlining the design and approach (8 minutes).

Example demo flow:

- User enters a medical research question
- System retrieves relevant research abstracts
- Multi-agent system evaluates research quality
- System generates a summarized research insight

Q&A with the panel (2 minutes).

Dataset Name

PubMed / MEDLINE Research Abstracts

Primary Link

<https://pubmed.ncbi.nlm.nih.gov/download/>

Alternative Links

<https://huggingface.co/datasets/ncbi/pubmed>

<https://www.kaggle.com/datasets/bonhart/pubmed-abstracts>

<https://pmc.ncbi.nlm.nih.gov/tools/textmining/>

<https://www.ncbi.nlm.nih.gov/research/bionlp/Data/>

Format

XML, JSON, CSV

Key Fields

PMID

title

abstract

authors

publication_date

MeSH_terms

keywords

Reason

Massive biomedical research corpus with rich metadata enabling semantic search, advanced filtering, and research trend analysis. The dataset also supports integration with medical ontologies such as MeSH for improved query understanding.